

ARAVIND RAM

KUBERNETES / CLOUD NATIVE ENGINEER

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PROFFESIONAL SUMMARY

Cloud Native engineer with 2+ years of hands-on Kubernetes platform engineering at scale. Designed and operated EKS clusters running 200+ nodes and 120 TB+ data workloads. Expert in Helm chart authoring, ArgoCD GitOps, custom Kubernetes operators, and production-grade cluster hardening with OPA Gatekeeper.

SKILLS

- Kubernetes : EKS, Karpenter, Helm (chart authoring, templating, umbrella charts), ArgoCD, Kustomize, Gatekeeper/OPA, ExternalDNS, ALB Ingress Controller, Storage Classes, CPU Manager, Custom Resource Definitions, Operator SDK
- Cloud Native Ecosystem: Prometheus, Grafana, AlertManager, Envoy, ZooKeeper, ClickHouse Operator, MongoDB Operator
- AWS: EKS, EC2, VPC, IAM Roles for Service Accounts (IRSA), Route53, ACM, S3, ASG, ECR
- IaC: Terraform (modules, workspaces, state management), GitHub Actions
- Languages: Python, Bash, YAML, Go

EXPERIENCE

FRESHWORKS, CHENNAI

2023- PRESENT

Kubernetes Platform Engineer

- Cluster Design & Operations
 - Provisioned and operated multi-tenant EKS clusters on AWS for ClickHouse analytics platform (120 TB, 200+ servers)
 - Terraform-automated full AWS stack: VPC, Security Groups, IAM roles, Route53, ACM, S3 buckets, Auto Scaling Groups
 - Deployed and managed: Karpenter for node lifecycle, Gatekeeper for OPA policy enforcement, ExternalDNS, ALB Controller, ClickHouse Operator, ZooKeeper
 - Migrated deployment system from Kustomize → Helm umbrella charts; standardized values-based multi-environment config
- GitOps & Release Automation
 - Operated ArgoCD as the sole deployment mechanism across all environments; enforced GitOps discipline
 - Built GitHub Actions pipelines: Helm schema validation, automated ECR image sync, release tagging, ArgoCD deployment triggers
 - Implemented remote migration framework using ClickHouse Remote Functions for customer data onboarding; Route53 cutover automation
- Operator Development
 - Led full migration from Percona MongoDB Operator → custom Community MongoDB Operator (CSMDB)
 - Rewrote reconciliation logic for sharding management, backup workflows, PITR, resource lifecycle
 - Built integration tests and E2E coverage; Jenkins CI pipeline for operator builds

- Performance Tuning
 - Benchmarked Kubernetes CPU Manager policies for ClickHouse; static CPU allocation reduced throttling by ~90%
 - Evaluated ClickHouse Lazy Materialization feature; improved SELECT query performance by ~40%
 - Implemented tiered storage (EBS → S3 via TTL policies + local EBS cache) reducing storage costs while maintaining query performance
 - Observability
 - Full Prometheus + Grafana + AlertManager deployment; custom PromQL rules and Slack alerting
 - Elastic Stack + Kibana for centralized log aggregation across cluster workloads
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PROJECTS

- Infrastructure Cost Dashboard — AWS Cost Explorer + Lambda + custom dashboarding for per-tenant spend
- ClickHouse MCP Server — AI-assisted infra operational tooling
- MongoDB Migration — 16 TB DocumentDB → self-hosted MongoDB on Kubernetes

EDUCATION

Bachelor of Science (BS) Degree in Data Science and Applications

Indian Institute of technology, Madras 2023 - 2027

COURSEWORK

CKAD: Certified Kubernetes Application Developer

LANGUAGES

Tamil - Native

English - Fluent